

# A ONE-TEST-SPACE CHARACTERIZATION FOR ARBITRARY FINITE-DIMENSIONAL SEQUENCES

RUN FA\_BANACH\_001

**Status.** `literature_implied_answer` (full arbitrary-sequence one-test-space, bounded-degree graph). This is not presented as a new theorem of the run. It records that a later finite-test-space-to-one-test-space proposition, combined with the finite graph test-space theorem in the source paper, gives the arbitrary-sequence case that was not known in the source's Section 3.

## SOURCE QUESTION

Ostrovskii's *Test-space characterizations of some classes of Banach spaces*, arXiv:1112.3086, defines test-spaces for a class  $\mathcal{P}$  of Banach spaces by the equivalence

$X \notin \mathcal{P} \iff$  the test-spaces admit uniformly bilipschitz embeddings into  $X$ .

Theorem T : Ost11+ of that paper states that for every sequence  $(X_m)_{m=1}^\infty$  of finite-dimensional Banach spaces there is a sequence  $(H_n)_{n=1}^\infty$  of finite connected unweighted graphs, each of maximum degree 3, such that for every Banach space  $Y$ ,

$(X_m)$  admit uniformly isomorphic embeddings into  $Y \iff (H_n)$  admit uniformly bilipschitz embeddings into  $Y$ .

In Section 3, after recalling one-test-space characterizations for super-reflexivity, the source paper says that it seeks analogous one-test-space characterizations for classes defined by excluded finite-dimensional subspaces, but that at that moment this was known only for increasing sequences, not for arbitrary sequences.

## SUPPORTING LITERATURE

Ostrovskii's later *Metric characterizations of some classes of Banach spaces*, arXiv:1411.3366, records the following elementary proposition in the discussion of one test-spaces. If  $(S_n)$  is a sequence of finite test-spaces for a class  $\mathcal{P}$  of Banach spaces containing all finite-dimensional Banach spaces, then there is a single metric space  $S$  which is a test-space for  $\mathcal{P}$ . Moreover, if the  $S_n$  are unweighted graphs, trees, or graphs with uniformly bounded degrees, then  $S$  may be required to have the same corresponding graph property.

The proof sketch constructs  $S$  by choosing vertices  $O_n \in S_n$  and joining  $O_n$  to  $O_{n+1}$  by long paths. The nontrivial direction embeds the  $S_n$ 's in parallel hyperplanes of the target Banach space, separated by the path lengths.

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**Literature-implied answer.** *Let  $(X_m)_{m=1}^\infty$  be an arbitrary sequence of finite-dimensional Banach spaces with  $\sup_m \dim X_m = \infty$ . Then there is a single metric space  $S$  such that, for every Banach space  $Y$ ,*

*$S$  admits a bilipschitz embedding into  $Y \iff (X_m)$  admit uniformly isomorphic embeddings into  $Y$ .*

*Furthermore,  $S$  can be chosen to be an unweighted graph with uniformly bounded degree.*

*Proof.* Let

$$\mathcal{P} = \{Y : (X_m) \text{ do not admit uniformly isomorphic embeddings into } Y\}.$$

Since  $\sup_m \dim X_m = \infty$ , every finite-dimensional Banach space belongs to  $\mathcal{P}$ : no finite-dimensional target can linearly contain all members of an unbounded-dimensional sequence.

Apply Theorem T : Ost11+ from arXiv:1112.3086 to obtain finite connected unweighted graphs  $(H_n)$ , with maximum degree 3, such that

$$Y \notin \mathcal{P} \iff (H_n) \text{ admit uniformly bilipschitz embeddings into } Y.$$

Thus  $(H_n)$  is a sequence of finite test-spaces for  $\mathcal{P}$ .

The proposition from arXiv:1411.3366 applies because  $\mathcal{P}$  contains all finite-dimensional Banach spaces. It yields a single metric space  $S$  which is a test-space for  $\mathcal{P}$ . Therefore, for every Banach space  $Y$ ,

$$S \text{ embeds bilipschitzly into } Y \iff Y \notin \mathcal{P} \iff (X_m) \text{ admit uniformly isomorphic embeddings into } Y.$$

Since the  $H_n$ 's are unweighted graphs of uniformly bounded degree, the same proposition permits  $S$  to be chosen as an unweighted graph with uniformly bounded degree.  $\square$

## SCOPE

This answers the source paper's arbitrary-sequence one-test-space gap under the same nontrivial unbounded-dimension convention used in the source paper's Remark R : InfDim. The supporting proposition gives a uniformly bounded-degree graph. This note does not separately establish a maximum-degree 3 refinement for the arbitrary-sequence one-graph construction.

## SEARCH AND PROVENANCE NOTES

The run indexes were searched for arXiv:1112.3086, the title of the source paper, one-test-space terminology, arbitrary finite-dimensional sequences, finite-dimensional Banach test-spaces, locally finite and coarse-disjoint union language, and Ostrovskii. No prior packet for this exact identification was found. Web queries on June 19, 2026 for exact combinations of 1112.3086, one-test-space, arbitrary sequence, finite-dimensional subspaces, and Ostrovskii found the source paper but no explicit separate answer. The decisive supporting proposition was located in the local parsed source of arXiv:1411.3366.